

The Social Radars

S5: David Kirtley, Founder & CEO, Helion Energy

Released January 7, 2026

Jessica: [\(00:00:00\)](#) Carolynn, today we are here with David Kirtley, the Founder and CEO of Helion Energy, making fusion power plants. Amazing. Welcome, David. Welcome, David.

David: [\(00:00:11\)](#) Thank you for having me.

Jessica: [\(00:00:12\)](#) I'm actually really excited because originally, I reached out to have you participate in one of our Founder Mode episodes at the YC retreat this summer. And unfortunately, you couldn't do that. So I said, well then, why don't you come on a full episode later on? So we get to talk to you for the full story?

David: [\(00:00:33\)](#) Let's do it. Let's talk about how to build a fusion business.

Carolynn: [\(00:00:36\)](#) Let's do it.

Jessica: [\(00:00:37\)](#) Fusion. Okay. Just for our audience, and me, of course, can you explain fusion, you know, fusion one zero one?

David: [\(00:00:47\)](#) Yeah. So fusion is the process that happens in stars where lightweight isotopes, so literally hydrogen and helium, the most abundant elements in the universe, under intense pressure and heat and temperature fused together joining forming heavier elements. It's actually the process how all matter is created in the universe. And we don't really harness it here routinely on Earth to make energy and or electricity yet. And so that's what Helion aims to do, is build generators that take those same processes that happen in stars and do it here on Earth to make low cost, safe baseload electricity.

Jessica: [\(00:01:29\)](#) So we're essentially looking at unlimited clean electricity sometime soon.

David: [\(00:01:35\)](#) That's the goal. Probably not free though. We do have a business to run. But that's the goal. That's what motivates the team at Helion is that, you know, every day we do fusion.

We take hydrogens and heliums, and then under really intense pressures and temperatures, we fuse them together, make electricity. I think, what the models show is that you can do fusion such that you could provide electricity lower cost than any other cost of electricity. Because the fuel is common in all water on Earth. And if you could build the machines effectively and efficiently and low cost, then the fusion itself, the power that you can produce is also really low cost.

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Carolynn: [\(00:02:17\)](#) Can I ask a dumb physics question? Just to does the does the act of fusing the molecules together, is that what releases the energy?

David: [\(00:02:24\)](#) So first, in physics, no question is a dumb question. Like Okay. Good. Like, what we're doing is harnessing the fundamental forces of the universe. And so understanding that and understanding the details of how that works is really critical for all of us day to day.

And, yes, so what happens is that in the depths of the atoms, you take two hydrogen atoms. In fact, we use a specific one called a deuterium, which is a heavy hydrogen. A hydrogen is made up of a proton and an electron. And every hydrogen deuterium has an extra neutron and a heavier particle. And then we take a helium three, which is another isotope that has two protons and a neutron.

And when you fuse those together, you make helium four. But that helium four and the proton it makes are actually lighter than the parts you started with. And this comes back to the old $E = mc^2$, actually, which is pretty fabulous, is that that change, we call that the mass deficit, is the energy released.

Carolynn: [\(00:03:29\)](#) Okay. Okay. Alright. That's good. Thank you for going back.

I just need to set the stage.

David: [\(00:03:35\)](#) He got into the weeds there. Yeah. So, fundamentally, what this means for us on a day to day is we put in this deuterium, this helium three fuel, and what we get out of it is helium four, which, by the way, is balloon helium. It's the normal helium we use in everyday life, and extra hydrogen. And those particles come out hot.

They come out with a lot more energy and electricity than you put into them. And so now you can take it's actually, like, a million times more energy comes out than you put in to the individual reactions. And so you can harness that and make a tremendous amount of output electricity.

Jessica: [\(00:04:10\)](#) Oh my god. This is, like, world changing, which is pretty exciting to be talking to you.

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David:
[\(00:04:17\)](#) I didn't say the hard part though. The hard part that's that's the easy part. The hard part is that, to get to the conditions where the fusion happens takes a 100,000,000 degrees and it takes a thousand atmospheres. So hotter temperature than the sun and higher pressure than almost everywhere on earth until you go to actually, it's the same pressure as the Marianas Trench, the lowest part of the ocean. And so you have to get to those pressures and the temperature of the sun all in one place.

And that's what's hard. That's what's taken physicists and engineers so long to be able to do this. And it wasn't until our last generation system, three years ago, that we were able to build a machine that did that, with our technology.

Carolynn:
[\(00:05:04\)](#) Well, we did this in the seventies, didn't we, in nuclear power plants? Or is or what are you doing something different than, like, those old seventies nuclear power plants?

David:
[\(00:05:12\)](#) Yeah. So you're thinking about nuclear fission. And so this is the

Carolynn:
[\(00:05:15\)](#) Exact separating.

David:
[\(00:05:16\)](#) This is the exact exactly right. Exact opposite where you're taking uranium, plutonium, heavy elements and Yeah. Cracking them in half and releasing the particles that are trapped inside.

Carolynn:
[\(00:05:27\)](#) Okay. But

Jessica:
[\(00:05:28\)](#) Part of fission is that there are these radioactive sort of byproducts. Right, David?

David:
[\(00:05:34\)](#) Yeah. The parts that are left behind, we call those the actinides. But the parts that are left behind are the, are the big radioactive byproducts in nuclear fission. And there's some other complications around potential meltdowns and energy release and all those things. That does just doesn't happen in fusion.

Okay. It's a it's a fundamentally different process.

Jessica:
[\(00:05:58\)](#) Okay. I'm gonna ask you to say one sentence again because it struck me. The thing about hotter than, like, the sun and also deeper than the deepest part of the ocean, say that sentence again.

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David: [\(00:06:11\)](#) That to get fusion to happen here on Earth, you have to heat the fuel to hotter than the temperature of the sun. In fact, 10 times the surface of the temperature of the sun. And at the same time, have it to be very high pressure. In our systems, it has to be higher pressure than the deepest part of the ocean on earth. So deeper than the Meryonic Trench.

Jessica: [\(00:06:32\)](#) Okay. That's gonna scare away almost every single startup founder I've ever met. So I wanna talk to you about how you I wanna back up and talk to you about how you got into this because what, what a big thing to be

David: [\(00:06:48\)](#) Challenged.

Jessica: [\(00:06:49\)](#) Attacking. Yeah. What a challenging thing to be attacking that no one succeeded in doing for years and years and years. So can we back up and tell us how you first started getting into this?

David: [\(00:07:01\)](#) Yeah. So I, was probably unique and, and this isn't that unique. I think that as a teenager, I said, wanna do something that changes the world. And, and I think I think everyone starts there, and then starts to look at the world with their lens of what are the problems I wanna go solve in the world. And for me, where I was at the time is I saw a lot of challenges with energy.

Clean water, and that comes down to energy. For a lot of my childhood, my father was in the Navy. And so we went from military base to military base all over the world. And I got to see lots of parts of the world, but also see that, you know, see the oil tankers, see I lived on a small island and see the desalination plants and see some of the other challenges we had and went to, I wanna go solve energy. And, you know, it always struck me as a child of an engineer and a and a scientist that most of the energy in the universe, in fact, matter too, is created from this process, this fusion process that happens in stars that we just don't do here.

And in fact, in some ways, it's a little it's a little bit it always felt silly to me that we use the after effects of fusion. We use the sunlight that, you know, a tiny fraction of the sunlight created that hits Earth, and we convert that to electricity. Or that gets converted into plants and then into eventually, you know, millions of years later into oil, and that's what we use. And it's like, aren't we using the actual process? And so that's what I went off to study.

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And I went and studied this for a number of years in my undergraduate academic education and an early master's work. And what I saw was that the fusion that we were doing, the approaches for fusion, were brilliant, run by brilliant scientists and engineers, that were focused on approaches to fusion that would eventually get us to fusion here on Earth. Like, I absolutely believe that. And I believe that, you know, the amount of work that those brilliant scientists, engineers have done over the since the nineteen thirties and forties has progressed the field of the physics and the engineering of fusion massively. But I also saw that some of the business and commercial aspects were missing.

That what we were seeing was it was going to take, we already knew this at the time, tens of billions or hundreds of billions of investment to prove a prototype. That that those technologies would never have a return on investment. Couldn't see how that amount of investment and a given amount of decades to build those machines would ever return low cost electricity. So you're not you're doing really great engineering and science, but you're not getting to the product. I also saw that and this is maybe a little selfish.

I like to build things. And the physicists and engineers that I was working with, we would be designing something, but probably never actually build it. It would never see it turn on in our career. And I was like, that that's not for me. Like, I wanna go build a thing that works, and I wanna see it.

I wanna participate in it. So I went off. I actually pivoted my career away from fusion and took the physics of high temperature gases. We call it a plasma, actually, and electromagnetics and all the things I learned. And I pivoted into space propulsion and started running teams and designing technologies to do to take some of that physics and engineering from fusion and apply it to rockets.

And in fact, some of the thruster technologies are what are the thruster technology that we developed when I was working at the Air Force Research Labs is the technologies used in Starlink right now to run those spacecraft that we developed.

Jessica: Really? That's
[\(00:10:52\)](#)

Carolynn: Very cool.
[\(00:10:54\)](#)

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David: [\(00:10:54\)](#) And shared out into the world, and then they took it and ran with it a million miles past where we were. But that told me, hey. We can actually take some of the same physics and engineering and modern technology and go way faster than what others had thought. 2008, I moved to Washington State to run a team to do this. And that team also took some of those technologies, and we applied it to fusion going backwards of, hey.

Why don't we try to develop systems that are much smaller than people have ever done before, that are much more efficient? They use fiber optics and high speed computing and high speed and, and modern power electronics that these other approaches that I learned in my academic programs didn't use, and now apply those modern technologies, essentially Moore's Law. Take Moore's Law and apply it to fusion. What can we do? And what we found is we could go way faster.

And so in the last ten years now, we've built seven prototypes that do fusion, where most folks are a decade or thirty years between prototypes. So that has enabled us to iterate quickly on the science, iterate quickly on the engineering, and then and then now build full scale systems. We're on our seventh generation system that's running every day.

Jessica: [\(00:12:08\)](#) And Polaris.

David: [\(00:12:09\)](#) We'll be running Polaris and doing fusion.

Jessica: [\(00:12:11\)](#) Wait. This is a good segue, actually. Because I asked Sam this week. I said you were coming on the show. So I asked Sam Altman because, by the way, I always look at, when we're interviewing someone, and I think Carolynn does too.

I look at the application just to see what was there. And you have a what I call a Sam special, meaning just your names and the company name was in it. So I said, I think Sam, like, hand recruited Helion. Is that true?

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David:

[\(00:12:42\)](#)

Yeah. I think it's I think it's pretty true that, when we met Sam, he and this is this is part of Sam's genius, is that he was thinking about the power problems of data centers ten years ago and longer now. And, you know, we were out there building systems that did fusion and still in the traditional model, the academic model of write a paper, write a proposal, submit it for a grant, win a grant, build a thing, learn a bunch of science and engineering about that thing, graduate a few students, and then and then rinse and repeat. But with the technology, it was moving much faster than other people. And so Sam actually came out and visited with a with a stack of textbooks, and we dug in for several days into the technology, the physics of fusion, and the business path and the and our iteration speed.

And I think that that resonated a lot. And then, he said, now I I'm you should go to Y Combinator. And because they're gonna teach you how to move fast and build a business that can actually scale.

Jessica:

[\(00:13:50\)](#)

So first, I wanna just read you what he said to me. Because I said, what do you remember about them? And he said, how earnestly they built relative to any other fusion company in the space. There were no fancy PowerPoints or anything, just building. Visiting their office, it felt like the YC approach to fusion versus how traditional VCs used to do it and hire MBA MBAs for it.

And I just I loved that description. And does that does that resonate with you?

David:

[\(00:14:21\)](#)

Yeah. I think it I it does that we were focused on and I didn't know this at the time, actually, that this was the this is how you build software that use software products that users want, is that you just build and iterate quickly and then deploy something, learn to collect data on how well it's working, and then iterate again. And that's what we were doing. And our goal was just to how do we get fusion to the world as fast as possible by every means necessary. And so when he visited, I think we had one machine operating, we had one machine half complete, and we had one machine and half disassembled.

And we're just iterating on all of those three in parallel.

Carolynn:

[\(00:15:06\)](#)

I ask you just what what's the size of these machines?

David:

[\(00:15:09\)](#)

So when Sam visited, we were in a, probably about a 5,000 square foot warehouse. The machine itself was it was a meter in diameter and about, 10 meters long, So about on the order of 30 feet long. And now they're a bit bigger. So we have all of Polaris is in a 27,000 square foot facility, and it's about 10 feet in diameter and about 40 feet long and a lot of power electronics that fill the rest that fill the rest of the building.

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Jessica: [\(00:15:43\)](#) I'm gonna just get this over with because I said to Paul, I think you flew up and visited Helion once. He came up with our son, I think, to visit you guys in Washington. And I said, what do you remember, Paul? And he said, he remembered how terrifying and gigantic the machines looked. Quote, palpably advanced.

So, Carolyn, I think we need to go on our field trip. Needs to stop by Everett, Washington. For sure.

David: [\(00:16:16\)](#) For sure. Yep. And I don't think he got to see, certainly, Polaris was not was not fully constructed yet and not operating. So he was seeing some of the previous systems.

Jessica: [\(00:16:26\)](#) Okay. So, Ken, you said quickly how you went to you moved to Washington with your team, but can we back up just really quickly? Do you remember what was going on when you first got started? Like, what were you focused on?

David: [\(00:16:40\)](#) It's so funny. I think that, you know, you look back on it and you can see the clear, pathway of how you ended up where you were. But at the time, maybe it just felt messy. Our goal was to build power plants as fast as possible. And what we saw was that the things that we were doing essentially, Sam met us, still, we were thinking about we were gonna do material science and plasma physics and fusion electricity and space propulsion and went to Y Combinator and spent time at Y Combinator.

And it was clear that you gotta focus. You gotta what is your product? And what are you gonna deploy in the world? And so that's it. And so we then narrowed down the vision of Helion to only that singular vision, making generators.

In fact, a very specific size of generators that where it can fit in the shipping container and move quickly and build toward build that.

Jessica: [\(00:17:38\)](#) Well, actually, that was another question I was gonna ask you. When Sam said when he visited you and said, now you need to do Y Combinator, I mean, no offense to us, but weren't you like, dude, we're building,

David: [\(00:17:50\)](#) You know, power plants

Jessica: [\(00:17:52\)](#) Here, generators. Like, like, what was your thought when he said that?

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David:

[\(00:17:56\)](#)

In all honesty, I was a little skeptical that we had been running a business for a while that, you know, collected lots of money from government grants. We build lots of hardware. You know, we had the basics of we had QuickBooks, we were doing the business parts, and we had incorporated. And we had done all of those parts, and we had a deck. It's interesting to hear Sam's version of our deck in ten years later, but we had a version of a pitch deck, and we thought we had that business, and we were moving faster than everybody else we knew.

And I was wrong that that, my, cofounder and I, Chris and I, spent the summer in the Bay Area at for Y Combinator. We would alternate, actually, where we would spend he would spend time while I was working on pushing building the machines. And then, and then I would spend time while he was up building the machines. And, and our other third cofounder was focused entirely on the hardware up in at that time in Redmond. And, yeah, what we saw at Y Combinator was just so, a different mindset of focus, of focus on what is the thing what how do you build a product that users want?

And for us, the user is everyone who buys power. And what they want is low cost, clean base load, and really safe electricity, and they wanted it yesterday. And so let's go focus on that and strip away any other side technology projects. So really laser focused on that one product. Two, focus on the business on really all the only the things that matter.

Don't focus on and maybe this is what Sam saw was the sort of the hints of this. But don't focus on the parts of the business, that aren't pushing the product forward. And so that meant things like we're no longer going to graduate as much students. Like, we're not gonna spend time training students anymore because that doesn't develop the actual technology as fast as focusing on just building. It means investing in the manufacturing upfront and building the capacity so that when you discover a new engineering or new science challenge at that time, you have the internal capacity to go build it and iterate and evolve it.

And then just streamline and focus. And every day, ask yourself, is this a thing that helps me deploy this product faster? And if it's not, then don't do it or push it for later.

Jessica:

[\(00:20:28\)](#)

That's interesting because focus is a key thing that Y Combinator brings to the table that I think the world doesn't realize how important focus is when you're building something, whether it's a consumer app or fusion company.

David:

[\(00:20:43\)](#)

And think about where you put your treasure. Where are you putting your resources? And are you investing your expensive investment capital in the parts that accelerate the business, or is it window dressing? Is it extra? Or does it slow down the business?

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Because you're preparing you're putting too much process in place. And so we think about that pretty constantly of does this accelerate the business? Does it accelerate product development? Does it accelerate the deployment? Does each hire accelerate it?

And really having that be the focus day to day, every hour of how you run the business.

Carolynn: Did you guys do demo day?
[\(00:21:21\)](#)

David: We did do demo day. It was a wild experience for me. So as a as a as a as a scientist, as an engineer who's given a ton of talks, I went in, to the YC partners with, like, this is what I think my pitch deck should look like. And it had a lot of words, like, just

Carolynn: Of course.
[\(00:21:42\)](#)

Jessica: Oh, I bet it did.
[\(00:21:43\)](#)

David: Yes. Annotations. They had a reference page where you're gonna, like, reference this. Like, it was very, very excellent scientific presentation, if I say so myself. But and they were like, no.

You need a picture. One picture. What's the vision? What are you actually developing? Mhmm.

Don't tell me about all the nuts and bolts about how you're gonna do it. You better understand that because you're gonna get questions on it. But tell me what the vision is. Tell me how the audience tell me how the world is gonna come along behind Fusion and Helion and deploy this. And so it was a it was a it was a pretty wild wake up call.

And the thing I found that was really interesting was that you learn your own business better. You learn your own product better. Because when you have all that detail to focus on, you are distracted. You are not focused. You are not accelerated.

And you have a honestly, a bunch of crutches that you could you could rely on rather than focusing on, like, what do we actually care about? We care about deploying power 1¢ a kilowatt hour, lower cost than anything else on the planet, and how do we get there? And what are the things to get us there? That was What was the picture,

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Jessica: By the way?
[\(00:22:56\)](#)

David: What
[\(00:22:57\)](#)

Jessica: Was the picture? The literal picture? Yeah. It was Carolynn, we would have been there.
[\(00:22:57\)](#) We would have this.

Carolynn: Yeah. And I was This actually sounds kind of familiar.
[\(00:23:03\)](#)

David: Yeah. So we had we had a couple of pictures because I wasn't able to down select, but
[\(00:23:06\)](#) we had but so still working on it. But a fundamental picture of the fusion itself. And this is something that differentiates Helion is we do fusion day to day. We do it.

And so when you do fusion, it's actually like a bright fuchsia branding of Helion. You see this bright fusion glow, this bright fuchsia background. And so it's like fusion is real. It's here. We do it.

This isn't science fiction anymore. This is a thing we do. So that was one of an actual operating fusion machine. The other was, what do you think the world looks like? Let's talk about the environment.

Let's talk about the impact of large scale clean electricity. And so there was some version of, like, clean water, and the environment. And so focusing on those things. And then be able to talk to the details of how you get to that one center kettle water. What are the tech technology paths you need to develop?

What are the proof points to be able to do that and get there?

Carolynn: And did you raise a lot of money?
[\(00:24:06\)](#)

Jessica: I was just gonna say, how did investors hide out? To this. Right.
[\(00:24:08\)](#)

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David:
[\(00:24:12\)](#)

We got a lot of interest, a lot of interest, and a lot of and a lot of offers at the time. In the end, what we found was that, and this gets back to that focus part, is that what we have done at Helion is be very careful around how big the cap table is and how many people are invested in Helion. And so we didn't actually end up taking offers that day or during demo day, and closing any closing around then. And so it wasn't until nine months later when we had worked with a number of those investors and then a bigger pool also and done diligence with them, make sure they really understand the technology, gotten them comfortable with the technology, but also getting us comfortable that they're comfortable with the technology because this is a bit different than what maybe people had seen at demo day before. So it was really important to us to, one, have a have a smaller pool of folks because we knew we were we would go on to, we raised a billion dollars now, multiple rounds as we built multiple prototypes and deployable fusion systems.

And then also make sure that everybody's on the same page about the timelines and, the amount of hardware and capital we're gonna have to build and deploy.

Jessica:
[\(00:25:34\)](#)

I'm sitting here with my mouth open a little bit because this is such a power move, David, to, like, sort of, like, you know, spend time, really get to know the investors, and sort of did you weed people out in the process saying, oh, they're just gonna sort of be impatient or cause problems? Or did it did you really get to know them?

David:
[\(00:25:55\)](#)

Yes. Yep. And that and that the round that we ended up building was of only a handful of larger investors, but also ones that were spending the qualification for me at the time, and it's a good question of whether we would make that decision now. So this is maybe a little bit of hindsight. Was that they were going to spend the time to either learn some of the engineering and physics or hire somebody that's gonna learn the engineering and physics and do technical diligence.

Interesting. Yeah. And for me, the thought process there I've never talked about this for what sort. This is super this is a really great question. Was that I knew that, we had a lot of big problems to solve and that, it was going to take iterations of both technology and machines as well as we're here in Everett Building, we went from 5,000 square feet to 500,000 square feet now of manufacturing.

And so and that was gonna take some real development and timeline and patience.

Jessica:
[\(00:27:01\)](#)

Yeah. Patience being the operative word here. Interesting. Who were can I ask who those investors were in that first sort of post demo day round? Was Sam in it or he had invested earlier?

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- David:
[\(00:27:15\)](#) So at that time, Sam was not. So Y Combinator was, so, our first round was led by Mithril Capital. Mhmm. And then and then we had
- Jessica:
[\(00:27:26\)](#) Oh, that's Peter Thiel. That's Peter Thiel. Right? Okay.
- David:
[\(00:27:29\)](#) And then we had Capricorn Investment Group, and Capricorn led our series b round. And we've already talked about publicly the other folks that that were in that. We had another half dozen folks, that were at Demo Day that had invested.
- Jessica:
[\(00:27:43\)](#) Okay. Moving on then. So it sounds like your time with Y Combinator was well spent.
- David:
[\(00:27:51\)](#) Oh, for me, for the business, fabulously well spent.
- Jessica:
[\(00:27:55\)](#) Okay.
- David:
[\(00:27:56\)](#) And so we have and we, even now, are, continue to as we bring in executives and leaders at Helion, how do we this comes back to Founder Mode too, which is why I'm sorry I missed you because I think that would have been a really fabulous time together.
- Jessica:
[\(00:28:12\)](#) Tell us quickly about your Founder Mode thing.
- David:
[\(00:28:15\)](#) We still as new executives and leaders come on, one, focusing on hiring to make sure I'm in and Chris and I are in the hiring process because we're looking to continue to grow that at the company. I think it's absolutely, critical that keeping that founder mode mindset, focusing on what matters, absolutely driving to that as hard as you can, taking no excuses for why things should be slower or we can't do a do that or solve that problem. And then getting in the weeds, as annoying as it is sometimes to the teams, Chris and I, George, the other founders, we are in the weeds. When there's a problem to be solved, we're gonna go and help solve that problem and not just hope it gets done. And so and we I want all my leaders to be that way.
- Jessica:
[\(00:29:06\)](#) So you did YC in summer fourteen. You know, you started a little bit before that. You've been doing this for a while. Did you have to train yourself on how to become a leader and a manage manager as your company grew?

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David:

[\(00:29:20\)](#)

Definitely, I have had to grow. Like, that is that is absolutely for sure. But part of it, I think, has also been I've had different development periods in my personal growth where I actually strayed away from Founder Mode a little bit and have come back. And so at the time that we joined Y Combinator, the founders at we're a we're a team. We are less than 10 people then, and we're well over 500 people now.

We Wow. Were very much, whether we knew the terminology at the time, in that mindset of very hands on, all very involved in the business, you know, literally hands on to the point of where we were turning wrenches and building machines, but also really paying attention to the details so that we could execute quickly and make the machines work and do fusion. And so putting all those pieces together, we were really focused on that, but we hadn't really thought about the framework. Anyway, we're just sort of naturally doing it. Over the years of building Helion, scaling, building now multiple machines that then set continual world records for fusion, that is sometimes drifted.

It it's easy to relax into more of a an a business mindset, more of a traditional business. Take your hands off the reins and say, like, I'm just gonna let this part of the business do its thing and not really be really, you know, focused on it. And then and then in some ways, we we've come to regret that, that there have been pieces where I look back and I'm like, woah. We've put bureaucracy and process. We slowed down.

We didn't accelerate. We slowed down. Why did we do that? And had to come back an adviser for many years who said, David, are you Eisenhower or Pat? Do you lead from behind the scenes orchestrating the field and then letting your generals go off and, you know, move the go forward?

Or do you get in the trenches? Do you get in the tank, get in the trenches, and then actually go and lead on the front lines? And then they and they spent many years telling me that we should I should be Eisenhower. I should be behind the scenes, set us the best set us specifications, hand it off, and let it just execute by itself and not pay attention to it. It turns out that's not me.

And it turns out that's none of the founders at Helion. That, it's it's get to the front lines and get in there and solve the problems. And then train folks, motivate them, give them the resources they need, and then let them go continue the thing because there's something over here, another battle to fight, another fire to put out, another manufacturing line to build, another hiring pipeline to open, another fundraiser to do to go and then and jump onto that and put all your all your effort into that that thing and solve that problem. And then build the team to go do that and then they can go run. And then and then go and build and build the business that way.

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And that turns out that one that resonates with me as a human, but it also has been, I have seen, the most effective. And I've been told it doesn't scale. People tell me that all the time, and I don't think that's right. I think that as long as what you're leaving behind when you go solve some part of the business is a part of the organization that is still operating in founder mode and going forward at velocity at high speed and hopefully accelerating, if you've if I've done a good job, then you can go off and focus on something else for a while. And then check-in every once in a while, see how it's going.

Carolynn: [\(00:32:59\)](#) Do you ever get tempted, to take this technology and work on something that's, like, easier and faster and lucrative? Because

Jessica: [\(00:33:09\)](#) Mhmm.

Carolynn: [\(00:33:10\)](#) Because it sounds like that's how kind of how you started until you really focused. And so I just wonder how much, like, temptation is there, maybe from yourself, but also from outside investors or whatever to, like, do the easy thing.

David: [\(00:33:21\)](#) There is constant external pressure to do that. A specific one is we use a fuel called helium three, a rare isotope of helium, that is safe and clean, but pretty rare. And we very early found and in fact, didn't invent this. This is one of the earliest ideas in fusion is using this fuel, which is most people, if you don't use this, they use a different fuel. And people don't use it because it's pretty rare and expensive.

And so what we developed very early is a way to make it, to actually make it here on Earth. And we've shown we can make it. We have now have a whole process to filter it and isolate it for ourselves, but we could go sell it. We could go sell it today. And it's, you know, total ball total addressable market, if you really squint at it, is maybe several \$100,000,000.

So it's it's a reasonable market, and it's and it's useful and would be good for humanity. In fact, it's used for medical imaging as a as a product, and it's not our goal. It's not the trillion dollar market. It's not clean, safe electricity for all of humanity. It is very niche.

But we could do it today and get and have customers that have come to us asking for it. And we say no. We say no because, one, it's not it doesn't solve the problem we want. And I worry that I've watched businesses get diverted by that early stage, much smaller, easier to grab product, and then stop there. And we don't wanna do that.

The other part is I also look at market dynamics, is that you look at how markets like that change. Yes. It's super lucrative now. Great. You go solve that problem.

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I drive costs lower, and now very quickly, it's not as lucrative. And I think we've had the freedom. This gets to something we were talking to earlier, but it gets we've had the freedom to say no because of some of the investors that we've picked, because of the work that

Jessica:
[\(00:35:27\)](#)

We're Yeah. Yeah.

David:
[\(00:35:28\)](#)

If they were interested in a three year return or even a five year return, then they would be pushing us very hard, and I may not be able to say no.

Carolynn:
[\(00:35:37\)](#)

Right. Right.

Jessica:
[\(00:35:37\)](#)

Yeah.

David:
[\(00:35:38\)](#)

By really focusing folks that really are looking towards that global scale deployment of clean electricity as the market, that that has given me the space as a leader and as the CEO to be able to focus the team on that. And it's a monthly conversation at Helena.

Jessica:
[\(00:35:57\)](#)

I bet. So you were saying that, you know, the idea of Fusion was started back in the thirties. They were researching it and did all this great work. And over the years, it's always been like, oh, Fusion's twenty years away. Twenty years away.

What has happened that now makes it not twenty years away? In fact, according to you, it's three years away, right, with this Microsoft deal? Like, what has changed to make that happen?

David:
[\(00:36:24\)](#)

Yeah. We broke ground on our power plant for Microsoft earlier this year. And so we are absolutely pushing forward and believe we can deploy electrons on the grid by 2028. And what's enabled that has been other technologies. There's been a tremendous amount of progress in fusion.

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Our understanding of the physics of those 100,000,000 degree gases today is not what it was twenty years ago and was is not what it was twenty years before. And so there's been a tremendous amount of that work that's being has been done. And I cannot never discount how great that has been and how great some of the government funding that supported that, the basic science, has been. But in parallel to that, Moore's Law happened. That the price of compute or the way we think about it is the amount of power that you can pack into a chip, into silicon, has so radically changed in the intervening, literally decades, Moore's Law that that has applied to us.

I think back to what the folks in the nineteen fifties and sixties did in fusion, and I am, like, in awe of that work because you had work with the computer didn't exist. So this was, like, hand calculations of really tough physics, things that we wouldn't even try to do by hand anymore. And they were able to build machines that did fusion by we haven't talked about the mechanisms of fusion too much, but large amounts of electricity. In fact, they are able to take our current Polaris right now is running at over a 100 gigawatts of peak electricity into the machine and large amounts of electricity coming out of the machine. And they were able to build systems before the transistor existed that could commute large amounts of current in very short periods of time, And it's with vacuum tubes.

It's absolutely amazing to me. And in the intervening time, silicon has now evolved to where we can I'll give you hard numbers here. We need to be able to turn on electricity in, we measure it in nanoseconds, in ten nanoseconds of time. We have to be able to and when I started my career, that was very hard to do. Ten nanoseconds is less than a gigahertz.

A gigahertz CPU is off the shelf. You can get it anywhere now. But it used to be really hard. We have the amount of data we have to collect for every pulse where we do fusion is measured in tens of gigabytes of data. And that hasn't changed.

That has always been the case. We've always needed that amount of data. But, again, when I started my career, we talked in kilobytes and megabytes, and that's just not the case anymore. It's we can we can actually handle that. And so we're not talking about gigabytes anymore.

We're talking about terabytes of data Yeah. That we collect at Helion. And that's that's things that we can do now. The other big one is fiber optics, is that in these systems, we have, in Polaris, for instance, we have tens of thousands of large scale semiconductors that are commuting, that are switching for the doing the electricity, and each one is controlled by a fiber optic. And that fiber optic has to respond again in nanoseconds of time scale.

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That used to be really, really hard, but you can go buy a 10 gigabit, So that's way less than a nanosecond fiber networking card from Best Buy today. And it's it has just totally transformed our ability to diagnose these systems, to understand what's happening inside of them, to control them, to run them. Because the physics has never changed. The gas, this high speed 100,000,000 degree gas has always moved so fast that we needed one nanosecond or tens of nanoseconds of time scale, and now we can do it. And also high speed computing just in itself, we have a small programmable logic brain on every single all distributed all over the machine.

We used to need big supercomputers to understand what was happening real time in these, but we don't anymore. We just put a little programmable logic all over the machine, and there's a large, you know, network bus, that that runs the whole thing over Ethernet. And that has now enabled us to do what we what was twenty years ago very, very hard and expensive to do, and twenty years before that was impossible to do.

Jessica: Wow. That is so cool. Tell us about your latest machine.

[\(00:40:47\)](#)

David: Our seventh generation machine is called Polaris. We named that after a star that does helium fusion, also happens to be the north star, the guiding star.

[\(00:40:51\)](#)

Carolynn: And Polaris.

[\(00:41:02\)](#)

David: And Polaris lives in a building we call Ursa, named after Ursa Minor, the little dipper. That's the constellation that Polaris lives in. That's how we name the plants, and we name the generators themselves that go inside the plants. We finished most of the mechanical construction last year, and we're able to begin making plasmas in December of last year for Polaris. We spent this year now hooking up more power, to Polaris, not just making these fusion plasmas, running at millions of degrees, but now superheating them to tens of millions of degrees.

[\(00:41:03\)](#)

And in Polaris, we are now have it's fully complete, and we've been running now at full power doing fusion for, for, several months now. And so now it's an optimization. It's tweaking. It's getting the power up. It's getting the power out that we're focused on.

Carolynn: I have a question. It sounds like it takes a ton of energy to make energy. This is sort of like that 50,000 foot. But it the goal here, unless I've really misunderstood this whole conversation, is that you make so much more in the output than you did in the what is the ratio?

[\(00:42:00\)](#)

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David:

[\(00:42:19\)](#)

What most folks in fusion are trying to do is make fusion reactions that are have tremendous, what we call yield, energy out per energy in. What we're trying to do is a little bit different. Our goal is to be very, very efficient with every all the electricity we put into it. And so this is what we proved, in 2014 during our Y Combinator time, is that we could take electricity, from the wall, put it into fusion electromagnets, the parts that do the work, and then recover 96% of that electricity Oh. Before you've done any fusion.

So now the fusion part only has to make up that last 4% and a little bit extra to sell. Most folks in fusion lose most of the energy they put in and most of the energy they pull out. So they're in the efficiency of maybe, if I'm being generous, 10% efficiency. Most a lot of folks are less than 1% efficiency. And what that means is the fusion process has to make up all that loss.

It has to be much more energetic. It has to be bigger, more complex, more expensive, take longer to build. And so by focusing on efficiency of it, of what do we care about? What's the product? It's not fusion.

It's electricity. And so I better be saving every elect electron I put in. I should pull out and save as much as I can. And so, therefore, our goal is now fusion, for us, we wanna make on the order of one and a half to two times the electricity we put in as output. Most people in fusion are, what, 30 times to 300 times the energy.

And so that that's the goal for what we do. And in Polaris right now, when we turned it on earlier this year, we're already recovering over that 90% of that that electricity that we are putting.

Carolynn:

[\(00:44:10\)](#)

That's amazing.

Jessica:

[\(00:44:12\)](#)

Wow. I feel like I've been sitting here with my mouth hanging open this whole conversation. I'm learning so much, David. I'm so excited. I feel like I'm gonna know just that a little bit more about fusion than I used to, which was not much at all.

So this is real. You signed this deal with Microsoft with I might add a penalty clause in

David:

[\(00:44:33\)](#)

It. Mhmm.

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Jessica: [\(00:44:35\)](#) Maybe you don't wanna answer this question, but if something were to go wrong between now and 2028, what do you think is gonna be the thing that's go goes wrong? Could you predict?

David: [\(00:44:47\)](#) Yeah. I mean, what I'll tell you is it's almost a thing you know, is that not everything is gonna work the way you think it will. And so what we like, that's that's the thing you really know when you're doing something that no one's ever done before, is there will be unexpected delays, whether they're business or they're engineering. There'll be new things you learn. Some will accelerate you.

Some will decelerate you. And so what we've done is we've built the business to solve problems, to be and that's that's part of Founder Mode. It's not saying, wow. There's a problem. I can't solve it.

But it's how do I solve it? Or I think that might be a problem. How do I invest today so that if it's a problem tomorrow, I will be able to solve it tomorrow? And so we've now that's how we've been successful of building six generations of machines is trying to see as far ahead as we can and plan for that. And so the things I worry about are the things a lot of the things we're talking about.

What if we're not 95% efficient, but we're 94% efficient? What if we're 93% efficient? How do we build the systems to be able to evolve that? What if, we haven't had this problem, and I've been very, fortunate for it, but what if we have delays in permitting? What if we have delays in actually doing the building?

There are financial penalties on the on a PPA. I think any real agreement that you have with your with your customer, and Microsoft's our customer, should have that. Like, like, we should be holding businesses accountable for delivering what they what they say. And so that was a requirement for us is that we wanted a customer that would that would that would have that as part of the contract. And so it's now building the systems in place.

Okay. We're 93% efficient. What do we do? Well, we've built now the manufacturing capacity that's bigger than what we needed for the bare minimum because we're gonna need some more power electronics. We're gonna need to do more fusion.

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We're gonna have to squeeze it a little bit harder. We're gonna have to get to a little bit higher temperature. So let's now build those systems already so that we can handle that and build the manufacturing and the engineering to be able to support that as we learn. And so a lot of it is starting to think as much as you can around what are the things that may not work out exactly as you as you planned so you can build the knobs to tweak that so you can adjust. And then keeping a company mindset that's problem solving based and excited and optimistic about solving problems.

And in some ways, that motivational piece is almost the harder piece than the manufacturing piece.

Carolynn:
[\(00:47:13\)](#)

So you just said, like, you're a problem solver. You iterate. You build. But sometimes there's problems that you have literally no control over, government, you know, regulatory, that kind of thing. Did you have any kind of setbacks where it's like, oh my gosh.

They're not gonna let us do this thing, and I have no workaround? Did you ever have a moment like that?

David:
[\(00:47:32\)](#)

I think when we founded the company, there were a few of those we were really, really worried about. The existential company killers that I don't have any that that I don't have much control over. And regulatory was one of those where, I mean, I think we Sam and I, I'm sure, talked about this ten years ago, which was even if we were able to solve all the technology problems and build those prototypes that we eventually did that did fusion, you might get regulated in a path of like, traditional nuclear fission power and not be able to deploy quickly, that it takes ten years to license a machine, that it takes a billion dollars in studies before you're ready to go build something, even though technology and the physics doesn't doesn't warrant that. And so we started investing in it. We started doing building the real honestly, building relationships, meeting those regulators.

I remember Yeah. 2018, I called the Washington Department of Health who handles who now is our regulator. But at the time, they regulated particle accelerators in hospitals. I said, you don't you don't know about fusion yet, but you're our regulator. Here is our proposal for our fifth generation machine and how I want you to regulate it, and here's a check.

Let's go start the process together. And it took years of us working with them to understand what drives them, then them to understand us of what are of the real safety concerns. This is big industrial power. There's things

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Carolynn: [\(00:49:04\)](#) Well, I was just gonna ask. We are all familiar with what the concerns are with nuclear power, and we actually talked about it, at the beginning. But what's the what's the thing that people are worried about with fusion? Like, what's the big disaster? Is it just an explosion because it's so heavy and hot and like, what is it?

David: [\(00:49:20\)](#) The biggest concerns we have in running Polaris today is mostly high voltage. It's the electronics. And so when we run these machines, we have large amounts of pulse power, large amounts of stored electricity that then very quickly, 100 gigawatts worth goes into the fusion machine. And then very quickly, large amounts of electricity come out of the machine. And so, we don't want humans in that environment.

And so that's the biggest one where we actually when we're running Polaris, no one's in the building in that 27,000 square feet. Those aren't the hardest ones, though. It's probably the biggest real safety concern. This is still an atomic process. You're fusing atoms together.

And much like a particle accelerator in a hospital, you do make ionizing radiation. X rays, you make neutrons. Those come off of the machine. And so we wanna shield humans from that. It's actually, an interesting story of when I went to the Department of Health, and I said, this is our shielding permit.

This is how we're gonna protect humans from the machines that are that are making, like, your particle accelerators that are making ionizing radiation. They said, great. Where is the patient? Because our calculations always have a patient inside the room with the machine. And I said, no patient.

They're outside the room. Like, we're gonna have just like in a hospital, we'll have what we call a vault. We'll have a concrete room with a door that closes, and you have operators on the outside, but we only have operators. And they said, oh, this radically changes this. We can go forward and go we can we can do this quickly.

And that's been really enabling for us is to be able to build systems we can monitor and operate those. But while you're running these machines, when you're doing fusion, you're not in the room with the fusion machine, just like you're not you don't other than the patient, you're not in the room with those big particle accelerators.

Carolynn: [\(00:51:07\)](#) Speaking of Moore's Law, you could use robots at some point, right, for if you needed

David: [\(00:51:12\)](#) Yep.

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Carolynn: You know, mechanical things, beings in the room with the with the with Polaris?
[\(00:51:13\)](#)

David: Yep. There's work being done all over the world on we call remote access and remote handling for using robots that do that.
[\(00:51:19\)](#)

Jessica: I have a question from Paul because I said that I was interviewing you. And I said, what question would you ask, David? And he said, I'm gonna read it. Presumably, you've already done simulations where you achieve net power. Can you manufacture things that are as perfect as the software?
[\(00:51:27\)](#)

What's the hardest thing to make as perfect as the software?

David: Oh.
[\(00:51:49\)](#)

Jessica: Does that make sense?
[\(00:51:49\)](#)

David: Yeah. That's a good question. That's a great question. There's sort of two classes of things I think about that are as hard to make as perfect as the software. One is that the physical machines themselves are large.
[\(00:51:50\)](#)

They're 40 feet long. And but the accuracy of some of the smallest parts, we measure in thousandths of an inch, one thousandth of an inch. And so you have to be able to manufacture components that are both can have the scale when they're all put together of 40 feet long and be as accurate as one thousandth of an inch. This is well bar better than one part in a million, and that that's hard to do. That's really, really hard to do, and it's taken us a lot of work to do that.

And early in Helion, it was really hard to do. We this is a really great question because we, have a technology we use now, which seems like magic to me sometimes, in that, laser trackers. That is that and this is some of the things like scanning LiDAR you might have for electric cars now. And you can build in Polaris, in the building of Polaris, we have little retro reflectors, little mirrors all over the building and all over the machine. And real time, we monitor.

We have laser trackers that can monitor the precision location within a thousandth of an inch over the whole building. And real time as we watch when it gets cold out or hot outside, the building moves and shifts, and we can detect that now. And that's been

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Jessica: Oh my god.
[\(00:53:27\)](#)

David: So enabling. When we built Trenta, our sixth generation machine, it took us almost a month to align all the parts. We put all the parts on large rails. We align the rails, then we had to tweak them in three dimensions. And we had people out every day with tape measures and even laser tape measures, adjusting it, little hammers, shims, all kinds of things.

And now we just put retroreflectors on everything, and we were able to align it in, let's say, hours, though it might have been faster than that. Wow. Wow. Versus weeks and a month. And it was, like in our schedule budget, I had budgeted a month to align this.

And they're like, David, it's done. And I'm like, what do you mean it's done? Like, it can't be done. Like, I'm gonna come down here, and I'm gonna, like, check the measurements because you just told me you accelerated from a month to an hour. And that's that's the power of some technology development or technology evolution.

The other is the same thing as a time scale is that we have the a hot gas moves quickly, and so you have to control it to this very high speed, one billionth of a second. And it's very easy to simulate one billionth of a second. This gigahertz computer can go do it. No problem. But doing it in real life is hard.

It's things like the fiber optic cable I mentioned is great, but depending on the length, the speed of light in that fiber optic cable has a given time. And so and that's longer than the time we care about. And so you have to now measure and understand the length of that fiber optic cable because the speed of light will limit you to a given time scale. And so doing it in real life in real space, the physical limits of the speed of light, the speed of electrons moving in wires, the turn on time of a real semiconductor versus a computer model. Those are a lot of the things that that that we spend a lot of time on is understanding those details and getting them right.

Some of the things that have gone the most wrong have been not getting those right. On Trenta, it took a month to align the machine. I didn't plan for a month then. On Polaris, I planned for a month because I had learned my lesson, but not on Trenta. And so that we just burned schedule on that because we didn't get it right.

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And on Polaris, we it went faster, which is great. But, similarly, on some of the details of timing and how do you get everything to be as accurate and time as possible, that has we've spent a lot of time on that, and, and it's been some of the some of the hardest challenges. In practice because you don't learn that lesson until you build the whole machine, until you have these large scale systems. The small scale on the bench works perfect. No problem.

Go on the big scale when now the fiber optics are much longer and you missed it. So you gotta go build that in the computer system to correct for that. We can do that, but that's that's been that's been some really interesting engineering challenges.

Carolynn: You have a lot on your plate, David.
[\(00:56:17\)](#)

Jessica: I know. Lot. Like, how do you do all this?
[\(00:56:19\)](#)

Carolynn: Yeah. This
[\(00:56:22\)](#)

David: Is a
[\(00:56:22\)](#)

Jessica: Lot. We're overwhelmed.
[\(00:56:22\)](#)

Carolynn: But I do wonder, one thing that I'm imagining probably isn't one of your 99 problems is competition. And not because there aren't other people, companies working on this, but I just get the feeling that that's not something that you lose sleep about at night. Is that right?
[\(00:56:24\)](#)

David: Generally, we don't worry about fusion competition, or even natural gas and other power system competition. I think the need is so great in the world, especially with AI data centers and the power need there in electric vehicles. And as we've gotten cheaper and cheaper electric vehicles and more and more of them, the need for clean base load electricity is so big that the competition is with ourselves. It's just how do we build this absolutely as fast as possible. We do think about it from the point of view of giving away the technology, sort of publishing schematics, and how we think about how we talk about it.
[\(00:56:42\)](#)

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But, really, at the end of the day, it comes back to how do we just build as absolutely fast as possible.

Carolynn: [\(00:57:26\)](#) Well, I guess vision, if it cleans up its image, could be a competitive technology. I don't know I don't know if that's true or not. But

David: [\(00:57:35\)](#) I think it will always come down to timelines. Our goal is to get to where we're building fusion on assembly lines, a gigafactory of fusion generators where they're coming off the assembly line. They're being then deployed very quickly as well. One of the things that I was surprised by that went better than expected, and because we were told by every expert that you couldn't do this is part of Founder Mode. We're told by experts, you can't do this.

It's not a thing that's ever been done, which is for Orion, our power plant, we were granted, a determination of non significance for the environmental permit. And this is, like, in the

Carolynn: [\(00:58:18\)](#) Such a legal word.

David: [\(00:58:19\)](#) In the weeds of how, environmental permits work, but it's a classification given to some solar farms. It's not even given to most of them because the environmental impact can be so big. And they said no. No nuclear power can ever have it. Fusion power is too new.

It'll never happen. We went through multiple years of working with the experts of these are the real concerns of fusion. These are the hazards of fusion. And they said, no. Like, this is your classification.

And what that means is we could start building today. We and we could break ground. And we got that determination. We got a lease approved and then had had we're moving earth within the hour of the lease being signed. Because of the scale of the impact of fusion is so low that you can move that fast.

And I don't know that that, uranium based nuclear power will ever be able to do that.

Carolynn: [\(00:59:16\)](#) Will ever do that. Yeah. So the goal is non significance, which is kind of funny when you think about it.

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- David:
[\(00:59:22\)](#) That the environmental impact is so low that we can do the study, deploy it. And then our goal is right now, we still deploy site by site. You say, I'm gonna go to this community. I'm gonna build a plant here. We you spend a year working with the community and the local, environmental agencies, the utility.
- You go multiple years. You now then go deploy a plant. Our goal is to change the metric here and make it not like building airports where you go and work at building the site, but like building airplanes. We're up in here in Everett and right by the big Boeing plants watching the planes come out of the building every day, And that's where you want fusion. If you were gonna deploy terawatts and trillions of dollars worth of power plants and generators, you have to do it in that way.
- So now we build machines right now on conveyor belts, behind me here in Everett. And, and that's that's that's weird for fusion.
- Jessica:
[\(01:00:18\)](#) David, what do you do to relax and have fun in your minimal amount of time that you have?
- David:
[\(01:00:25\)](#) I have a nine year old son who's into ice hockey now, joined an ice hockey team. So I have become one of those parents at the at the I love my Saturday ice hockey games, and so that that's fun. And then and then a lot of, you know, trying to take some of the engineering, the science, the iterative development things that I have done and bring that home. It's Halloween season. It's my favorite season.
- And so, it's a lot of how do you build an, animatronics, and how do you take some of that engineering, the computer programming, and bring that home. And then so lots of fun projects along those lines.
- Carolynn:
[\(01:01:04\)](#) Alright. Go ahead. Does your
- Jessica:
[\(01:01:05\)](#) Son like those?
- David:
[\(01:01:07\)](#) Mostly. Mostly. Sometimes it's, cringe is the word that I've I've I've learned.
- Carolynn:
[\(01:01:14\)](#) Yeah. Familiar with it.

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Jessica: [\(01:01:15\)](#) Yeah. It only gets worse, David. It only gets worse. The thing about this interview to me that feels so profound is that I feel like, you know, generations will look back on this weird time in history when power was expensive and limited. And, man, David, if Helion solves this, it's like as big as the industrial revolution kind of thing.

I mean, do you ever step back from all the things you have going on and think about that?

David: [\(01:01:50\)](#) I do. When we even when we were in Y Combinator and the business case for Helion and the problem that we were trying to solve was big. It was fossil fuels. We're still growing. We're still, powering more coal, using more coal than we ever have before for humans, and we are gonna go solve that.

We are gonna go solve that. We are gonna go replace fossil fuel plants. But at the same time, power usage in The United States, at least, was growing at 2% per year. Not a rapidly growing market. And in fact, if you look back at the cost of electricity leveled to inflation, the cost of electricity was decreasing then, ten years ago.

That changed. That changed with the massive growth in the EV market and now large scale data centers. We are now seeing not 2% growth per year in electricity infrastructure and deployment in The United States, but 20%. We're no longer seeing prices go down a few percent per year. They're radically increasing all over the world, including The United States.

And so the problem is much bigger than what we are the problem we are trying to solve now is much more pressing than it was when we founded the company. And so to me, the pressure to solve the problem, to deploy absolutely as fast as possible at scale, and not just one power plant. But I guess what I say to the team sometimes is if we demonstrate that you can do fusion at low cost and efficiently, and that's all we do, then we have failed. If we deploy one power plant that puts electricity on the grid from fusion for the first time, historic achievement, we've failed. We have to be at a place where we have built a product and a business that can build generators, gigawatts per day of deployed power at scale, and deploy it in the world and create both the business, the technology, the regulatory environment to be able to go do that.

That's when we'll have succeeded because the world needs that. And the demand is increasing, and it's increasing at a rate that we haven't seen before. And I think that I worry folks are underestimating the growth and the pressure we're gonna find for energy and electricity specifically.

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Carolynn: You're selling this power to utilities, in at least in the residential sense. Is that right?
[\(01:04:09\)](#)

David: So this gets into some of the really interesting, nerdiness around the power utility market and how all electricity is done that for Orion, our PPA with Microsoft is to deploy 50 megawatts of electricity to the grid. And it will be to Microsoft, but through utility. In this case, the Chelan the Chelan public utility, the PUD. And so it'll be electrons that go on the grid, and then Microsoft will be able to pay us through a power marketer constellation. But what we're also working with is our second customer, Nucor, who's a large steel manufacturer, to not do that.

But in fact, to put the generator right on their site, 500 megawatts, about 10 times as big, right on their site and give them power directly. That one's called behind the meter. And that's something that we don't really do industrial scale. Almost always, it's a large central generating station that goes to the grid on a transmission line and then goes to a data center. But our goal is to be able to put generators on the data center, on the manufacturing, on the factory sites and provide them power directly.

It's much more efficient and you can use some really cool engineering things around tailoring how you provide power, not for the grid, but you're providing it for a data center which wants different kinds of power. It wants DC. It wants higher voltage. It wants intermittent loads. It and so be able to dial that to the data center needs or to the steel manufacturing facility's needs, it would be really powerful from an engineering point.

We're pushing that as fast as we can.

Carolynn: That makes total sense. But like I said, for residential, unless you're gonna build something that can actually be attached to someone's, you know, garage, Like, we're still all gonna get this power from our utility. Is that correct?
[\(01:05:52\)](#)

David: For residential power, I think that's absolutely right. Fusion fundamentally is still large scale industrial power. We're gonna make them absolutely as small as we can, and they're still gonna be big. It's still gonna be a 25,000 square foot building, and it's gonna have a substation, and it's gonna be industrial scale power. And so we want modular where you have one generator per town rather, but it'll still be at that industrial scale.
[\(01:06:04\)](#)

At the scale that we're trying to deploy, I think that's okay, that that that we want to be able to deploy throughout the world, as fast as possible at large scale. And so some of those large industrial facilities is are gonna be required.

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Carolynn: [\(01:06:41\)](#) And that infrastructure is already built. Right? That infrastructure is already built. It already works. So, like, that does seem like the most efficient way to get this energy to the world, like, is your goal.

David: [\(01:06:51\)](#) I think that's the goal. What we are looking for is the new stuff that comes online. That new AI data center doesn't require lots of new transmission lines. So now you can take Fusion, put it on-site there, and power that data center directly rather than having to build a bunch of transmission lines also, which is

Carolynn: [\(01:07:10\)](#) A Right.

David: [\(01:07:10\)](#) Is which is a real tough problem.

Carolynn: [\(01:07:12\)](#) Right.

Jessica: [\(01:07:13\)](#) I mean, AI companies are gonna go bananas when fusion actually works for them. I mean, because they the cost of running their companies are so expensive. I mean, are do you get people calling you up saying, hey. How's it going? What's what's the timeline looking like?

David: [\(01:07:33\)](#) Yeah. When can you deploy? Because I'm we're ready for you. And so that's, for instance, having Microsoft as a as a first customer. That's a lot of the driver there.

And we were working with Microsoft since 2015. Almost right after Y Combinator, we started working with them where they were visiting us, watching our prototypes in development. We were working with their experts. They have a whole nuclear power and electricity teams, shockingly. Probably not shockingly since they're

Carolynn: [\(01:08:00\)](#) Not shockingly. Yeah.

David: [\(01:08:02\)](#) Where they have they deploy massive amounts of power. And as we built our systems and proved that they did work the way we expected them to.

Jessica: [\(01:08:10\)](#) This is just such a long game. I'm like I'm I'm just so excited to be on the sidelines, like, you know, cheering for you and learning more and watching this, and I cannot wait to see how this all plays out. I feel so optimistic after talking to you, David.

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David:
[\(01:08:28\)](#) And next time you are on a on a field trip and you wanna come visit and see a working fusion system, come out.

Carolynn:
[\(01:08:36\)](#) I do. Do too. Working fusion system. I'm

Jessica:
[\(01:08:39\)](#) I I'm not joking. Carolynn and I are gonna go on the road, and we're gonna stop by. And we're gonna see, Polaris and everything there. I can't wait.

David:
[\(01:08:50\)](#) Perfect.

Carolynn:
[\(01:08:51\)](#) Gonna be great.

Jessica:
[\(01:08:52\)](#) Thank you so much for coming on the show today. I feel like our listeners are gonna this is gonna be such a treat for them to have heard all of this story. I know I learned a lot, so I'm totally psyched we got to have this long form conversation with you today.

Carolynn:
[\(01:09:07\)](#) Yeah. Much better than fifteen minutes. Too much to cover here in fifteen minutes.

Jessica:
[\(01:09:11\)](#) Awesome, David. Best of luck over the next, few years, and I just I cannot wait to see what happens with Healian.

David:
[\(01:09:19\)](#) Thanks for having me.

Carolynn:
[\(01:09:20\)](#) Thank you, David. Thanks.

Jessica:
[\(01:09:22\)](#) Alright. Bye. Bye. Carolynn, that was so interesting.

Carolynn:
[\(01:09:28\)](#) That was complicated, and there were definitely some very complicated scientific concepts. But for the most part, I think the way David explains it, pretty digestible and pretty understandable. And it's actually so cool to learn stuff like this. I know.

Jessica:
[\(01:09:39\)](#) I feel like I'm smarter than I was this morning when I woke up. Yes. I certainly understand fusion better. Yes. So do I.

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I love that conversation. And it is such a long game that he's playing.

Carolynn: Yeah. I kept thinking about, like, this is like pharmaceutical. This is like biotech where there's just you have to you have to be in it for the long haul because drug development takes forever, and so does nuclear fusion.
[\(01:09:54\)](#)

Jessica: I mean, it is it is world changing if this stuff works, which he's proven that it can work.
[\(01:10:08\)](#)

Carolynn: Yeah.
[\(01:10:15\)](#)

Jessica: It's just a matter of deploying it, like, commercially. Right?
[\(01:10:15\)](#)

Carolynn: Scaling and
[\(01:10:18\)](#)

Jessica: Deploying. Yes. Scaling and deploying.
[\(01:10:19\)](#)

Carolynn: And I thought it was really interesting that he's building that or he has he's manufacturing in house his own helium that he could sell. Part of me, I wanted to ask, although I just maybe it's just a whole such a tangent, but, like, why not license that to people since it's so desirable? But I'm sure that, you know, it's probably like, well, that will help that will distract him. It's it's losing focus, and he was so adamant that focus is so important, which obviously is.
[\(01:10:21\)](#)

Jessica: Love how Y Combinator taught them how important focus was and forced them to focus.
[\(01:10:47\)](#)

Carolynn: Yeah. No.
[\(01:10:53\)](#)

Jessica: It's great. You know? Because that makes me feel like we add value to the hardest of hard tech. Yep. I definitely would like on our little field trip when we go to see we're gonna go see Stoke.
[\(01:10:54\)](#)

Right? We're maybe gonna see Charm. We're gonna go see Boom. But we're gonna we gotta go on a little field trip. Social Radars on the road, field trip to these things

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Carolynn: Okay. And
[\(01:11:18\)](#)

Jessica: See it. Because I think you saw it, you'd come away with your mind blown even more than hearing the description because these are Sure. This is a big operation.
[\(01:11:19\)](#)

Carolynn: Yeah. No kidding. Yeah. I would love to do that. That's a great idea.
[\(01:11:28\)](#)

Jessica: I can't wait for that episode to come out. It's gonna be a great one.
[\(01:11:31\)](#)

Carolynn: It's gonna be a great one.
[\(01:11:34\)](#)

Jessica: I'll I'll see you on the next one.
[\(01:11:36\)](#)

Carolynn: Okay. Bye.
[\(01:11:38\)](#)

Jessica: Okay. Bye.
[\(01:11:39\)](#)